



Arrow Ballistic Simulator

Ballistic Simulator for Archers

Introduction and program guide

v9.0 — 2025/2026

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1. Why this simulator

It is well known that all "projectile launches" arrows, javelins, cannonballs, kicked balls — theoretically follow a simple parabolic trajectory. The deviation from the ideal parabola depends on dozens of technical and environmental factors: this is the origin of ballistics, a discipline with extensive literature.

In the specific field of archery, scientific output is far more limited than for firearms. Studies on arrow ballistics have existed since antiquity but have often remained within the realm of practical experience, given the complexity of an arrow's trajectory — a flexible, non-rigid body with fletching — compared to that of a rifled projectile. Today's leading researchers are engineers and physicists who often work in collaboration with equipment manufacturers or with Olympic federations. (eg. [Western Hunter study](#), [OutLab — Fisica dell'arco](#)).

For the modest practitioner of this noble sport, all these studies — though rich in valuable information — are of little immediate practical use. The idea therefore arose to create an application that, by entering one's own arrow, bow and biometric parameters, allows the user to:

- **visualize** the real trajectory of one's own arrows, with attention to drop and residual energy;
- obtain a practical answer to the fundamental question: **with my bow and arrows, to hit a target at "X" meters and "Y" elevation, at what angle to the horizontal must I set my shot?**
- generate a **sight table** with projections on the riser, exportable to PDF and CSV.

- generate the **stringwalking table** for those who practice this technique: for each target distance, how many millimeters below the nock do you need to go with the grip on the string to hit the target.

2. The program

The code is developed entirely in **HTML + JavaScript** pure, with no server or installed framework dependencies. It is comparable to a single-page mini-website: simply click the link <https://arrow-sim.netlify.app> it with any modern browser (Chrome, Firefox, Edge, Safari). It works on PC, tablet, and Android and iOS smartphones.

2.1 Multilingualism and measurement systems

Version 9 is fully internationalized. In the program header you will find:

- **Language selection** (top-left corner): IT Italiano · GB English · ES Español · FR Français · DE Deutsch
- **Measurement system** (top-right corner, next to the theme toggle 🌞 / 🌙): m = metric (m, cm, m/s, J) · yd = imperial (yd, in, ft/s, ft·lb)

All input parameters and results convert automatically when switching system. Internal calculations are always in SI.

3. The physics of the simulator

3.1 Trajectory and numerical integration

Arrow flight is simulated by integrating the equations of motion with an adaptive 4th-order Runge-Kutta (RK4), with automatic step control via Richardson extrapolation (tolerance $\sim 10^{-6}$ m). This guarantees high precision even for steep angles or slow arrows.

The aerodynamic drag is modelled with a Reynolds-number-dependent C_D coefficient, with four modes: laminar (low Re), turbulent (high Re), auto (Re-dependent transition), and transition scenario (mixed boundary layer). Reference data come from JAXA/MSBS wind tunnel measurements (Ortiz Enriquez doctoral thesis, UEC Tokyo, 2021).

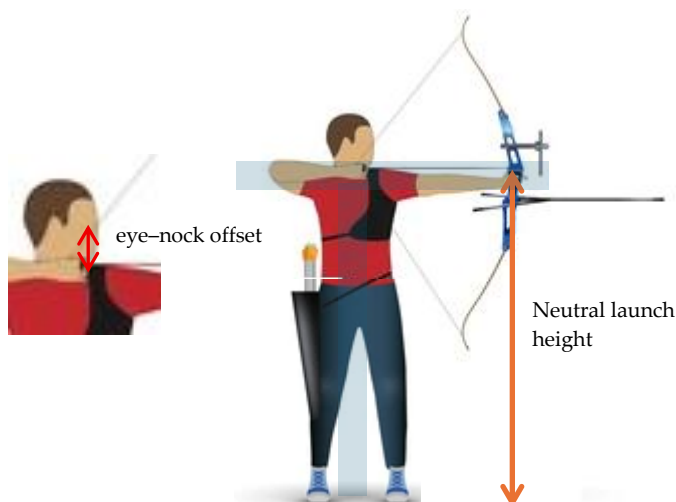
3.2 Modeled physical effects

Effect	Implementation
Air density	CIPM-2007 formula (temperature, pressure, humidity, altitude)
Altitude correction	ISA thermal gradient for pressure and temperature
Wind	Forces on the arrow's cross-sectional profile
Nocking point offset	Conversion mm offset → angular deviation via arrow geometry (arctan)
FOC influence	Drag penalty for FOC > 10% (stability/drag trade-off)
Postural correction	Iterative update of launch height during T-form rotation
Sight projection	Full eye–nock–riser geometry with parallax

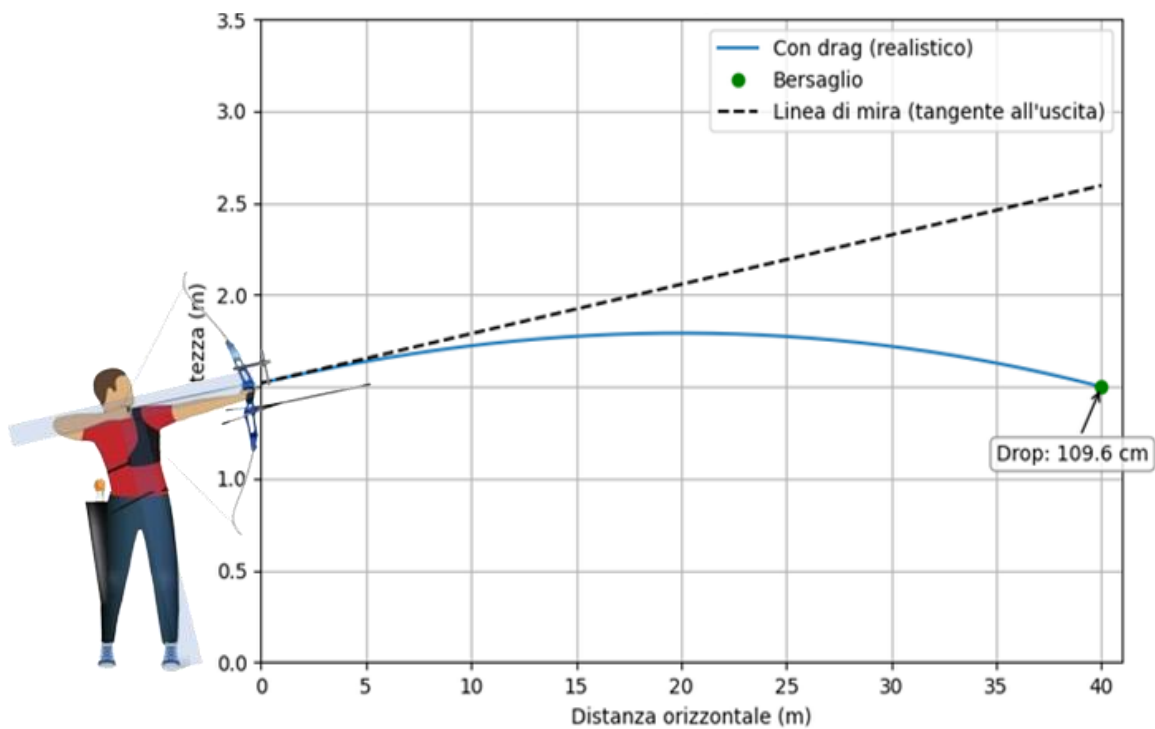
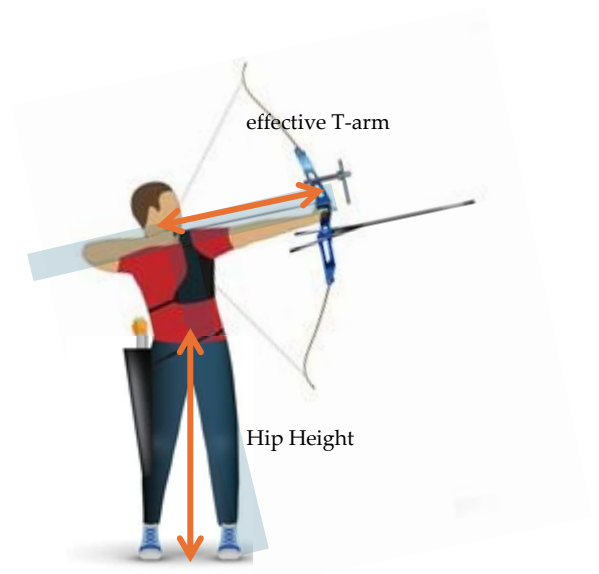
4. Archer biometry

The simulator considers the archer in the "T-position" with the arrow horizontal. The relevant biometric parameters are:

- **Neutral launch height:** height of the arrow tip with horizontal posture (depends on the archer's height and grip)
- **Hip height:** rotation centre of the torso during T-form rotation (approximately at the navel).
- **Shoulder-to-anchor length:** effective T-arm, from shoulder to anchor point.



With the **Postural correction flag** active, the simulator iteratively calculates the effective launch height of the arrow as the shooting angle changes when the T is tilted up or down, the



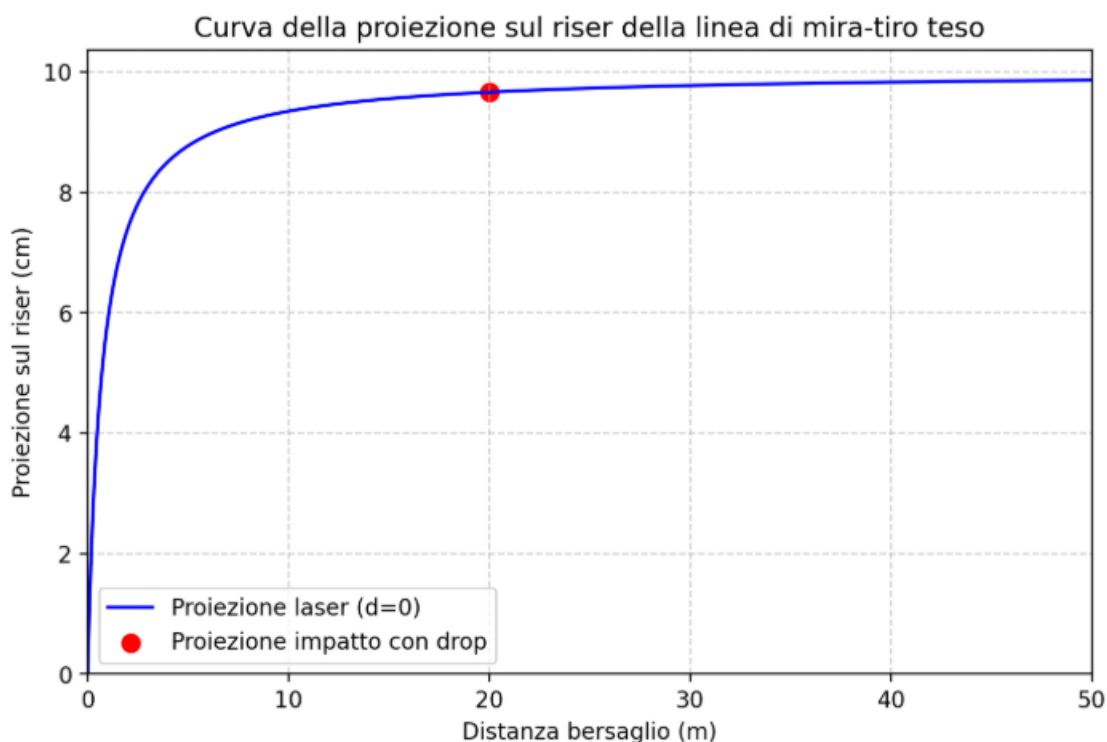
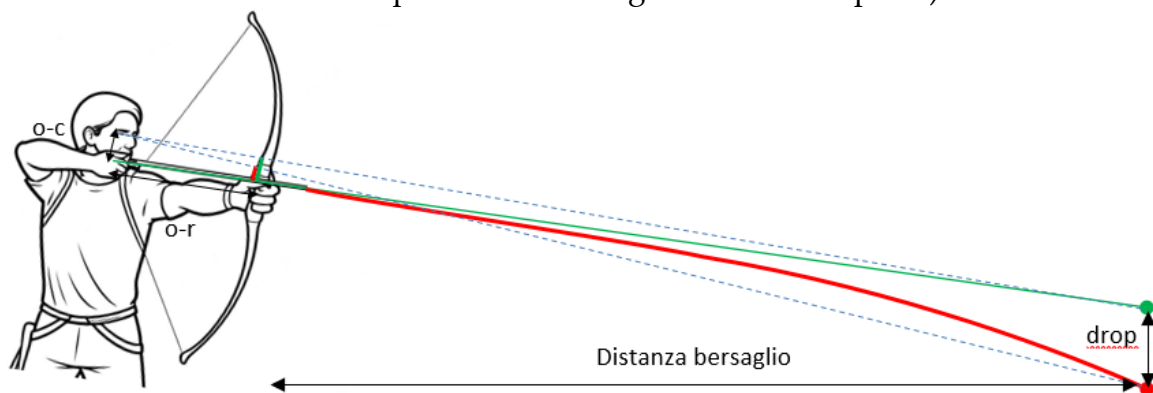
arrow departs from a slightly different height than the neutral one, which affects the trajectory.

5. Sight calculation

5.1 Projection onto the riser

Once the optimal angle to reach the target has been found via applied ballistics, the practical question is: how to achieve it? The simulator offers a geometric approach: imagine that a **laser beam** exits from the arrow on the rest. Where would we see it projected on the riser — in second vision — when we aim at the target at distance X ?

It is a matter of trigonometry (eye–nock–riser parallax). The projection is a function of three quantities: eye–nock distance, nock–riser distance, and arrow drop. For distances beyond 20–30 m the projection practically converges to the parallax distance eye–nock (which in the simulator corresponds to a fixed green reference point)



5.2 Nocking point offset

The Nocking offset field (always in mm, regardless of the selected measurement system) allows modelling of the vertical offset of the nocking point relative to the perpendicular to the string. The – and + buttons increase or decrement by 1 mm for convenience on mobile devices. Internally the value is converted into angular deviation via arctangent and subtracted from the calculated aiming angle.

5.3 Stringwalking

Stringwalking is a barebow technique in which the finger's position on the string varies depending on the distance to the target: moving the grip downwards produces a "lift" in the arrow, allowing you to hit farther targets without changing your visual aiming point.

Since version 9, the simulator has included the 🏹 Stringwalking tab, which automatically generates the grip point table for each configured distance.

Stringwalking is in fact (with a good approximation) equivalent to dynamically varying the nocking offset: moving the grip downwards a few millimeters/centimeters on the string produces the same "lift" in the arrow as would be obtained by lowering the nocking point by the same amount. The simulator exploits this equivalence: for each distance, it finds the optimal launch angle using RK4 integration, then calculates the grip point with the formula: $\text{gripMm} = \tan(\text{launch_angle}) \times (\text{arrow_length} \times 0.9) \times 1000$ where $\text{arrow_length} \times 0.9$ is the approximate distance between the rest and the nock.

In other words: by aiming the arrowhead at the target, the archer physically tilts the entire system (bow + arrow), and the grip point is the position on the string that automatically produces the right angle of elevation.

The table shows each target distance:

- **Hook (mm):** millimeters below the nock. Positive value = descending onto the string. Negative value = correcting with aiming (rare, only for very close distances).
- **Aim Angle (°):** launch angle required to hit the target.
- **E (J):** residual kinetic energy at impact.
- **V (m/s):** residual velocity at impact.

Practical Notes:

- The 0 mm reference corresponds to the hook immediately below the nock. Each archer can adjust the values to their own personal reference by subtracting the offset of their usual grip.

- The Eye-Nock field: for stringwalking, set 2-3 cm (eye-to-index finger distance); for the traditional Mediterranean grip, the typical value is 10-15 cm. This field only affects the construction of the Sight tab, not the calculations of the Stringwalking tab.

5.4 PDF export and calibration rulers

The  **Export PDF** button generates an A4 page containing:

- the complete **sight table** (drop, riser projection, energy, speed for each distance);
- the 1:1 **visual sight scale** with distance marks;
- two 5 cm **calibration rulers** (vertical and horizontal) to verify that the print is executed at 100% without automatic scaling — a common fault with many printers. Overlay a physical ruler to check.

6. Model limitations

The simulator is an educational and practical tool, not a certified professional system. The main simplifications adopted:

- **2D trajectory:** no lateral component (spine effects, crosswind drift)
- **Rigid-body arrow:** shaft flexion and the archer's paradox are not modelled.
- **Simplified bow efficiency:** constant efficiency is used, not the real load curve.
- **No Magnus Effect:** negligible below ~1,500 RPM (typical of field shooting)
- **Spine and stability:** influence C_d in a simplified way (multiplicative factor).
- **Simplified stringwalking:** the model does not consider the effects of spine or the residual vertical component of the resultant string forces. For typical field shooting angles, the approximation is negligible.

7. How to use it in practice

Tips to get the most from the simulator:

- **Chronograph:** if a measured speed is available, enter it as v_0 . Simulation accuracy will considerably be improved.
- **Optimal FOC:** keep FOC between 10–15% for the best stability/speed trade-off (automatically calculated from the balance point).
- **Shot up or down:** set negative heights to simulate downhill shots (from a hide or platform); adjust launch height and hip height to simulate kneeling shots.
- **Variable conditions:** update temperature, pressure and wind for more accurate simulations in different weather conditions.

- **Sight PDF:** always verify the print with a physical ruler on the two 5 cm calibration rulers before transferring the marks to the riser.

8. Availability and updates

The simulator is available as an open-source HTML file on GitHub:

github.com/CP52/Arrow-ballistic-simulator-for-Archers.

How to use it:

Online (recommended)

Open directly in your browser: <https://arrow-sim.netlify.app>

Works on PC, Mac, iPhone, Android and tablet — no installation required.

Offline — standalone version

Download the file:

`arrow_ballistic_simulator_v9_i18n_offline.html`

Open it with any modern browser (Chrome, Firefox, Safari, Edge).

Works completely offline from the very first use — no internet connection needed.

Ideal for field use, via AirDrop, Files or USB drive.

Base version (for developers)

The file “`arrow_ballistic_simulator_v9_i18n.html`” is the main source.

Requires an internet connection on first launch to download React and jsPDF from CDN. After the first load it works offline.

iPhone / iPad

Option 1 (recommended): open <https://arrow-sim.netlify.app> in Safari →

tap Share → ****Add to Home Screen**** for a dedicated icon.

Option 2: download ‘`arrow_ballistic_simulator_v9_i18n_offline.html`’,

open it in Safari from Files — works without internet.

*Released for personal and educational use, without warranties. Attribution appreciated. The code is developed by
Cesare Pagura with Claude (Anthropic), Padua 2025/2026.*

Enjoy and shoot straight! 

Cesare

Padova, 2025/2026